IterDANet: Iterative Intra-Domain Adaptation for Semantic Segmentation of Remote Sensing Images

Yuxiang Cai, Yingchun Yang, Yongheng Shang, *Member, IEEE*, Zhenqian Chen, Zhengwei Shen, and Jianwei Yin

***Abstract*— When segmenting the continuous proliferation of unlabeled remotely sensed images, unsupervised domain adaptation (UDA) has become one of the most critical techniques and achieved significant performance. But, in fact, there still exists a large performance gap between the existing UDA frameworks and supervised learning methods, for the majority of UDA frameworks do not consider the intra-domain gap in the target domain. In this article, to further minimize the complex intra-domain shift within the target domain in remote sensing, we propose a novel iterative intra-domain adaptation framework (IterDANet), which conducts inter-domain adaptation (InterDA), entropy-based ranking (ER), and iterative intra-domain adaptation (IntraDA). Specifically, first, to enhance the performance of InterDA built upon generative adversarial network (GAN)-based image-to-image (I2I) translation, we propose a new generator selection strategy (GSS) to assess and choose a well-trained generator for the inter-domain classifier. Then, to produce more accurate pseudolabels for IntraDA, we propose a new pseudolabel generation strategy (PLGS) to remove both high-entropy and low-confident pixels in predicted maps of inter-domain classifier. Finally, to better reduce the intra-domain gap, we propose to cluster all the target images into multiple subdomains using ER and iteratively align the cleanest subdomain with other noisy subdomains. The extensive experiments on the benchmark dataset, which includes cross-city aerial images, highlight the superiority and effectiveness of our IterDANet against the state-of-the-art UDA frameworks.**

***Index Terms*— Adversarial learning, image-to-image (I2I) translation, inter-domain adaptation (InterDA), intra-domain adaptation (IntraDA), remote sensing images, self-supervised learning (SSL), unsupervised domain adaptation (UDA).**

I. Introduction

WITH the continuous proliferation of remote sensing images, remotely sensed semantic segmentation, a pixel-level image interpretation task, has become one of the most meaningful and interesting research branches. For it is of great importance to numerous practical uses in remote sensing community, for instance, land cover monitoring, environmental and urban management, agriculture, climate change assessment, and so on. Recently, convolutional neural networks (CNN)-based models [1], [2], [3], [4], [5] have widely adopted to remotely sensed semantic segmentation and achieved impressive performances [6]. However, these CNN-based approaches, which are trained in a supervised manner, usually require a large amount of annotated labels for satisfactory performance. It is laborious and time-consuming to annotate the large-scale remote sensing images in pixel level. If we directly employ the classifier trained on the labeled images (source domain) to segment the unlabeled images obtained from different cities or sensors (target domain), the segmentation accuracy of classifier would decrease significantly, which results from the domain shift between the target and source domains [7], [8], [9].

In remote sensing, the distribution discrepancy between the target and source domains is generally caused by a variety of factors [10] during the image acquisition process. For example, the illumination, imaging times, and seasons may influence the color distributions of the images. Various images acquisition locations may lead to the intraclass variability; for example, the buildings of the different countries or cities may have various structures and architectural styles. In addition, the images acquired from different sensors may compose of various kinds of spectral bands; e.g., some consist of blue, green, and red bands, while others include infrared, green, and red bands. Besides the abovementioned factors, atmospheric conditions, imaging angles, and other factors would also result in the distribution discrepancy between images.

To mitigate the aforesaid domain shift without extra manual labeling, unsupervised domain adaptation (UDA) has been presented to apply the knowledge of labeled source domain to unlabeled target domain and has shown outstanding performance in cross-domain semantic segmentation. Overall, the majority of existing UDA approaches build upon three techniques: image-to-image (I2I) translation, adversarial learning, and self-supervised learning (SSL). In general, the UDA frameworks based on I2I align the data distributions of source images with that of target images in the feature or image space. By training the I2I network between the target and source images based on CycleGAN [11], Benjdira *et al.* [12] transformed the source images to target-like images, which have similar appearances and

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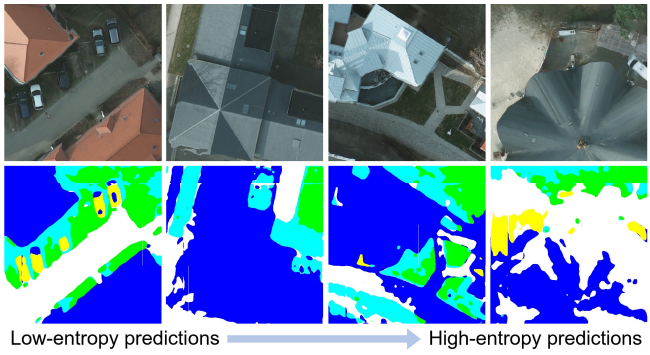

Fig. 1. Sample of intra-domain gap problem. (Top row) Four target images obtained from the same city, with diverse building styles and the roof colors. (Bottom row) Corresponding predicted results of Deeplab-V2. Some predictions are relatively accurate and low entropy (left), while others are high entropy and noisy (right), even if all the predicted results come from the same classifier.

data distributions to real target images. Then, the target segmentation model was trained on the target-like images with corresponding labels for original images. Besides the alignment in image level, Hoffman *et al.* [13] conducted I2I with matching the target and source images in feature and pixel level, which further minimizes the domain gap between the source and target domains. Without I2I, the UDA methods based on adversarial learning employed the discriminators to align the source domain with target domain in output or feature space of the classifier. For example, Zhang *et al.* [14] trained a discriminator to discriminate the features of source images from that of target images, while the classifier learned to fool the discriminator and extract domain-invariant features. Tsai *et al.* [15] proposed to align the output space of segmentation model instead of feature space, which leads to further improvements of performance. Due to lack of annotations for the target images, pseudolabels have been introduced to the UDA frameworks to improve the performance of target classifiers. On the basis of the UDA framework based on I2I network, Li *et al.* [16] added an SSL module, which utilizes high-confident target predictions as pseudolabels to further train the target segmentation model. It illustrates that the SSL methods can be added into the UDA frameworks to further enhance the performance of segmentation models.

Although the UDA frameworks have made great advents in the cross-domain semantic segmentation, there are still large performance gaps between UDA frameworks and supervised learning methods. We conduct extensive experiments and find when applying the same classifier trained with UDA framework [12] to segment all the target images, the classifier is able to achieve similar performance to the supervised learning method on some target images, but the performance degrades heavily on other target images due to the intra-domain gap within the target domain, as shown in Fig. 1. In general, the intra-domain gap is usually caused by many factors, such as diverse scene distributions, illumination, intraclass variation, and so on. For instance, as shown in Fig. 1, the buildings in the target images acquired from the same city have various roof colors and building styles, which results in the distribution shift between different images. To handle this problem, Pan *et al.* [17] proposed to conduct intra-domain adaptation (IntraDA) on the basis of inter-domain adaptation (InterDA) in three steps to reduce the intra-domain shift within the natural images. First, they trained the target segmentation model by reducing the inter-domain shift between the source and target domains via an adversarial learning approach [18]. Then, they clustered the target domain into clean split and noisy split based on the entropy of predicted results for target images. Finally, the clean split with pseudolabels was aligned with noisy split to mitigate the intra-domain shift by an adversarial learning approach [18], which effectively eliminates the intra-domain gap in the natural image datasets.

However, the problem of IntraDA in field of remote sensing raises some challenges. First, the majority of UDA frameworks do not handle the intra-domain gap in the target domain. Only rare UDA frameworks attempt to alleviate the intra-domain shift by clustering the target domain into two subdomains, but there may still exist distribution shift within the subdomains, which would degrade the performance of segmentation model. In other words, the existing UDA methods are inadequate to handle the complex and diverse intra-domain shift within the target domain in remote sensing, and the complex intra-domain gap within the target domain needs to be further minimized to narrow the performance gap between the UDA methods and the supervised methods. Second, the IntraDA builds upon InterDA and ER, which means poor performance of the InterDA and ER will lead to the failure of IntraDA. Although the existing I2I-based InterDA frameworks have achieved remarkable performances, the generative adversarial network (GAN)-based I2I models suffer from mode collapse and instability issues and may produce low-quality pseudoimages with noises, which would degrade the performance of the inter-domain segmentation model. Third, when producing the pseudolabels for IntraDA, the existing IntraDA methods directly utilize the predictions of the inter-domain classifier as the pseudolabels, but there are still some incorrect pixels in the pseudolabels, which would mislead the intra-domain segmentation model during the self-training process.

In this article, to deal with the abovementioned issues, we propose a novel iterative intra-domain adaptation framework (IterDANet), which composes of InterDA module, entropy-based ranking (ER) module, and IntraDA module. These three modules work as a whole, and all of them are indispensable for solving the intra-domain gap. Specifically, the InterDA module, which builds on the GAN-based I2I network, aims to eliminate the inter-domain shift between the target and source domains. To cope with the instability and mode collapse issues of the GAN-based I2I network, we propose a novel practical assessment approach, named generator selection strategy (GSS), to assess all intermediate generators of the GAN-based I2I during the training phase and select a well-trained generator, which can generate high-quality transformed images without noises and artifacts. Then, to produce more accurate pseudolabels and remove the wrong pixels in the predictions as many as possible, we propose a new pseudolabel generation strategy (PLGS), which takes advantages of the combination of softmax probability and entropy as the confidence assessment and filters out both

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high-entropy and low-confident pixels in the predicted results of the inter-domain segmentation model. Finally, to handle the complex and diverse distribution shift within the target domain, we propose a novel iterative IntraDA algorithm. The algorithm splits the target domain into multiple subdomains on the basis of ER and InterDA and then applies the cleanest subdomain with pseudolabels to iteratively optimize the intra-domain classifier on the rest subdomains.

In general, the contributions of this article are as follows.

1) We propose a novel IterDANet for cross-domain semantic segmentation of remote sensing images. IterDANet clusters the diverse target domain into multiple subdomains on the basis of InterDA and ER and iteratively reduces the intra-domain gap in remote sensing images. For all we know, IterDANet is the first study on unsupervised iterative IntraDA for remotely sensed semantic segmentation.
2) We propose a new GSS, which is a practical assessment approach, to reduce the impact of the instability issue of the GAN-based I2I translation and enhance the performance of InterDA, which is the basis of IntraDA. The proposed GSS can assess all intermediate image generators during the training phase and select a well-trained generator for I2I-based InterDA.
3) We propose a new PLGS to further improve the performance of IntraDA module and SSL methods. The proposed PLGS takes advantages of the combination of softmax probability and entropy as the confidence assessment and removes both high-entropy and low-confident predictions to produce more accurate pseudolabels.

II. Related Work

A. I2I Translation

The existing UDA models based on the I2I [19], [20], [21], [22], [23] usually transform the labeled source images x_s to target-like images $G(x_s)$ via GAN-based architectures, which train the discriminator D to distinguish the target-like images $G(x_s)$ from real target images x_t and encourage the generator G to produce verisimilar target-like images $G(x_s)$ to fool the discriminator D. Such learning process can be formulated as follows:

$$\mathcal{L}_{\text{I2I}}(G, D) = \left[\log D(x_t)\right] + \left[\log(1 - D(G(x_s)))\right]. \quad (1)$$

The target-like images $G(x_s)$ have similar data distributions or features to target domain, and the semantic contents of $G(x_s)$ are the same as the original source images x_s. Then, the target segmentation model can be trained on the target-like images $G(x_s)$ with corresponding labels y_s for the original images x_s [24], [25], [26], [27]. For instance, Bousmalis *et al.* [28] proposed to generate the target-like images by learning the distributions of the target domain in the pixel space, which shows impressive results. On this basis, [13] and [29] alternatively transformed the source images to target domain by aligning the images in feature and image space and further maintained the semantic contents during the I2I. In the field of aerial images, Benjdira *et al.* [12] utilized CycleGAN [11] to perform I2I between aerial images acquired from different cities and imaging sensors and then trained the target segmentation model on the transformed images generated by CycleGAN. Similarly, Tasar *et al.* [30], [31] proposed DAugNet and ColorMapGAN, which build on I2I, to reduce the domain shift between satellite images obtained from various cities.

B. Adversarial Learning

To align the source domain with target domain, some UDA frameworks adapt the segmentation model via adversarial learning in the output or feature space [32], [33], [34], [35]. These methods train the discriminator D to distinguish the features or output $F(x_s)$ of the source images x_s from that $F(x_t)$ of target images x_t and encourage the classifier F to extract domain-invariant features or output to fool the discriminator D, which can be expressed as follows:

$$\mathcal{L}_{\text{ADV}}(F, D) = \left[\log D(F(x_s))\right] + \left[\log(1 - D(F(x_t)))\right]. \quad (2)$$

For instance, Zhang *et al.* [14] trained a discriminator to discriminate the features of source images from that of target images, while the classifier learned to fool the discriminator and extract domain-invariant features. Instead of aligning the feature distributions, Tsai *et al.* [15] proposed to align the output of segmentation model, which takes the source and target images as input. On this basis, Vu *et al.* [18] aligned the entropy maps of the predictions of the segmentation model, which further mitigates the domain shift. In remote sensing, Al Rahhal *et al.* [36] trained the classifier via adversarial learning and adapted the classifier to align the labeled source domains with unlabeled target domain, which composes of aerial images obtained from different locations.

C. Self-Supervised Learning

SSL or self-training [37], [38], [39] has achieved remarkable progress in UDA [40], [41], [42]. These methods usually train the segmentation model F to produce pseudolabels $\hat{y}_t = F(x_t)$ for the target images x_t; then, the pseudolabels $\hat{y}_t$ and the target images x_t are utilized to train the segmentation model F by minimizing the classification loss, e.g., cross-entropy loss, which can be written as follows:

$$\mathcal{L}_{\text{SSL}}(F) = -\hat{y}_t \log F(x_t) \quad (3)$$

where the pseudolabels $\hat{y}_t = F(x_t)$ are the key to the SSL methods; hence, these methods seek to improve the accuracy of the pseudolabels. For example, when producing the pseudolabels, [16] and [43] removed the low-confident pixels in the predicted maps by defining a threshold for softmax probability. Similarly, to employ high-confident pseudolabels in self-training, Zheng and Yang [44] rectified the high-uncertainty pseudolabels by estimating the uncertainty of pseudolabels. Besides the confidence and softmax probability, an entropy of predicted results has been adopted to evaluate the pseudolabels as well, and Saporta *et al.* [45] selected the low-entropy predicted results to train the segmentation model with self-training.

All the abovementioned UDA approaches only learn to minimize the inter-domain shift between the target and source

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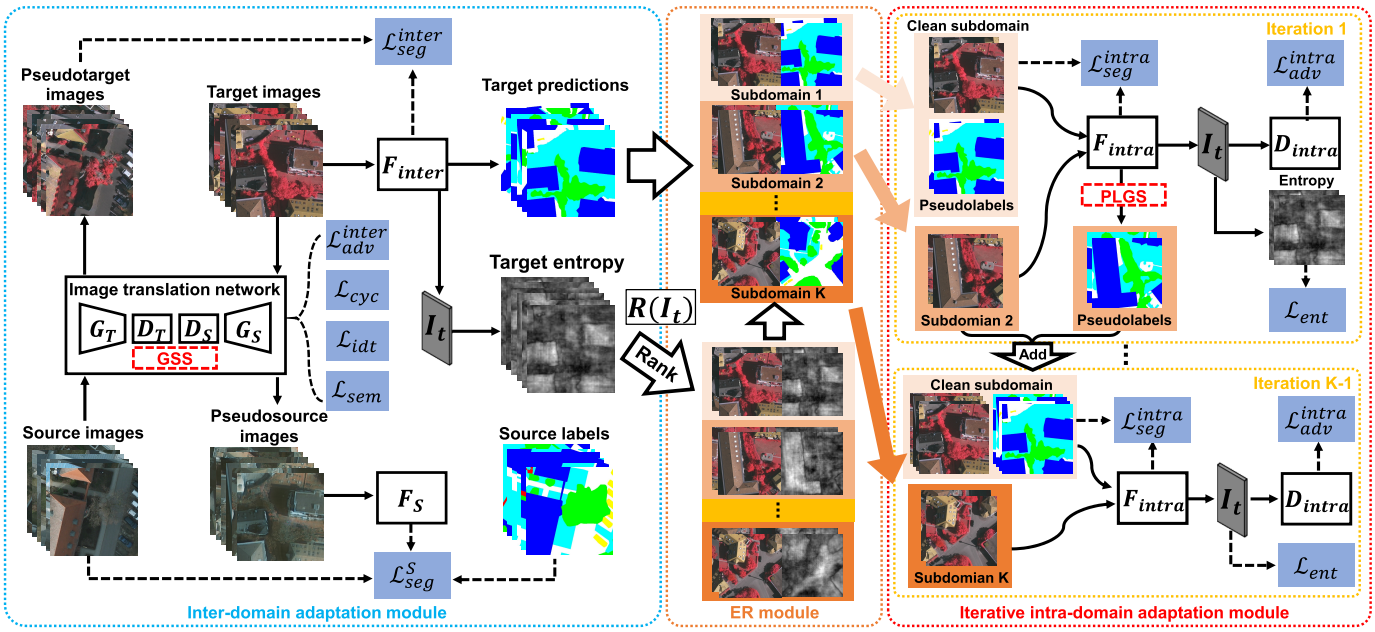

Fig. 2. Overall training process of our IterDANet. First, the InterDA module trains the inter-domain classifier F_{inter} and the source classifier F_S to reduce the domain shift between the source and target domains based on I2I network, which composes of two image generators $\{G_T, G_S\}$ and two discriminators $\{D_S, D_T\}$ alone with the proposed GSS. Then, the ER module ranks the mean entropy of the predicted maps generated by inter-domain classifier F_{inter} and splits the target domain into multiple subdomains. Finally, the iterative IntraDA module trains the intra-domain classifier F_{intra} and discriminator D_{intra} to align the clean subdomain, which includes pseudolabels generated by the proposed PLGS, with other high-entropy subdomains via the proposed iterative IntraDA algorithm. At test stage, the trained intra-domain classifier F_{intra} can be directly utilized to segment the target images.

domains, without considering the intra-domain shift in the target domain. As a result, there still exist many unsatisfactory predictions for the target domain.

D. Intra-Domain Adaptation

In remote sensing, the target domain in UDA is usually complex and diverse because of the large intraclass variability. To handle the intraclass variation, Sun *et al.* [46] proposed to cluster the objects into several subdivisions based on the features extracted by two orthogonal GANs (O-GANs) [47] and then trained a segmentation model for each subdivision. At test stage, the multiple subdivisions would be combined. Although this approach improves the performance of segmentation model, it is trained in a supervised manner and not suitable for the unlabeled target domain in UDA. For UDA, Pan *et al.* [17] first reduced the inter-domain shift by aligning the output space of source and target domains with adversarial learning methods, as illustrated in (2). Then, they clustered the target domain into clean split and noisy split based on the entropy of predicted results for target images. Finally, the clean split with pseudolabels was aligned with noisy split to mitigate the intra-domain shift. However, this method has two limitations. First, it directly utilizes the predictions of the clean split as pseudolabels, but there are still many wrong pixels in the clean predicted maps, which may mislead the segmentation model during training. Second, for the complex and diverse target domain in remote sensing, only clustering the target domain into two splits is not competent.

III. Method

In this section, we formalize the learning problem as the typical UDA task, where unlabeled target sample $x_t \in \mathbb{R}^{H\times W\times 3}$ and labeled source sample $x_s \in \mathbb{R}^{H\times W\times 3}$ with C-class segmentation label $y_s \in (1, C)^{H\times W\times C}$ are provided. Let G and D be the image generator and discriminator, respectively. F denotes the segmentation model, which takes sample x as input to output softmax probability $P \in (0, 1)^{H\times W\times C}$ and entropy map $I \in (0, 1)^{H\times W}$ for sample x. We aim to train an iterative IntraDA framework for cross-domain remotely sensed semantic segmentation as shown in Fig. 2. The proposed framework is composed of three modules. The first component is the InterDA module based on I2I [12], [48], which includes the proposed GSS to improve the performance of inter-domain classifier. Second, the target images are clustered into multiple subdomains based on the entropy of predicted maps, which are generated by the inter-domain segmentation model. In the third iterative IntraDA module, we utilize the proposed PLGS to produce more accurate pseudolabels, and the clean subdomain is iteratively optimized and aligned with other noisy subdomains via the proposed iterative IntraDA algorithm.

A. Inter-Domain Adaptation Module

The InterDA module aims at minimizing the inter-domain gap between the source and target domains and obtaining a well-trained inter-domain segmentation model for the ER module and the iterative IntraDA module. The existing I2I-based UDA methods have shown their power and robustness in the general UDA tasks. Thus, we utilize the bidirectional UDA framework [48], which can better reduce the domain shift caused by differences in spectral bands, instead of aligning the output space by adversarial learning.

1) Bidirectional Domain Adaptation: Following the bidirectional UDA framework proposed in [48], we apply two segmentation models F_{inter} and F_S to segment the images

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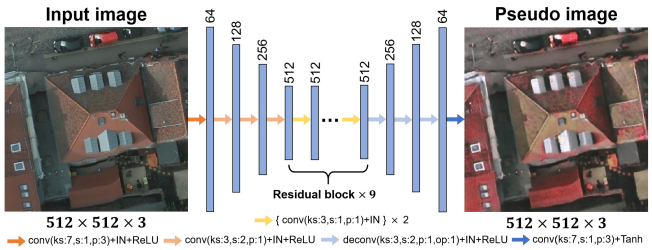

Fig. 3. Structure of generators G_T and G_S. ReLU, IN, p, op, s, and ks denote rectified linear unit, instance normalization, padding, output padding, stride, and kernel size, respectively. Conv and deconv stand for convolution and deconvolution.

drawn from the target and source domains. For the bidirectional I2I network based on GAN [49], two generators G_T, G_S (see Fig. 3) and two discriminators D_S, D_T (see Fig. 4) are utilized. When performing I2I between the source and target domains, we hope the data distributions of pseudotarget images $G_T(x_s)$ and pseudosource images $G_S(x_t)$ should be similar to that of the target and source domains, respectively. Therefore, we introduce the bidirectional adversarial loss, which encourages the discriminators to distinguish the pseudoimages from the real images and enforces the generators to generate verisimilar pseudoimages. The abovementioned bidirectional adversarial loss $\mathcal{L}_{\text{adv}}^{\text{inter}}$ for the two discriminators D_S, D_T and two generators G_T, G_S can be formalized in the following equation:

$$\begin{aligned}\mathcal{L}_{\text{adv}}^{\text{inter}}&(G_T, G_S, D_T, D_S)\\ &= -\sum_{h,w}\left[\log D_S\left(x_s^{(h,w)}\right)\right] + \left[\log\left(1 - D_S\left(G_S(x_t)^{(h,w)}\right)\right)\right]\\ &\quad + \left[\log D_T\left(x_t^{(h,w)}\right)\right] + \left[\log\left(1 - D_T\left(G_T(x_s)^{(h,w)}\right)\right)\right]. \quad (4)\end{aligned}$$

Besides the data distributions should be similar, the pseudoimages should be consistent with the original images semantically; otherwise, the mismatches between pseudoimages and labels of original images would degrade the performance of the classifier. Therefore, we apply three semantic loss functions to better enforce the generators to maintain the semantic contents when producing pseudoimages.

First of all, we hope that transferring the pseudoimages back to the original domain would not change the semantic contents of the images. Hence, to minimize the differences between x_s and $G_S(G_T(x_s))$, and x_t and $G_T(G_S(x_t))$, we use cycle-consistency loss $\mathcal{L}_{cyc}$ [11], which can be formulated as follows:

$$\begin{aligned}\mathcal{L}_{cyc}(G_T, G_S) &= \sum_{h,w}\left[\left\|G_S(G_T(x_s))^{(h,w)} - x_s^{(h,w)}\right\|_1\right]\\ &\quad + \left[\left\|G_T(G_S(x_t))^{(h,w)} - x_t^{(h,w)}\right\|_1\right]. \quad (5)\end{aligned}$$

Then, we hope that the generators G_T and G_S will not change the semantic contents of x_t and x_s, respectively. Therefore, to encourage $G_T(x_t)$ and $G_S(x_s)$ to maintain the original semantic contents with x_t and x_s, respectively, we adopt identity consistency loss $\mathcal{L}_{\text{idt}}$, which is defined as follows:

$$\begin{aligned}\mathcal{L}_{\text{idt}}(G_T, G_S) &= \sum_{h,w}\left[\left\|G_T(x_t)^{(h,w)} - x_t^{(h,w)}\right\|_1\right]\\ &\quad + \left[\left\|G_S(x_s)^{(h,w)} - x_s^{(h,w)}\right\|_1\right]. \quad (6)\end{aligned}$$

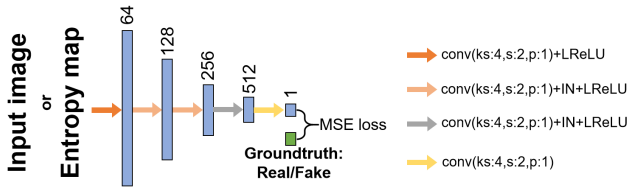

Fig. 4. Structure of the discriminators D_S, D_T, and D_{intra}. LReLU and MSE denote leaky ReLU and mean-squared error.

Finally, we hope that $G_S(x_t)$ and $G_T(x_s)$ should be semantically consistent with the original images x_t and x_s, so we utilize the bidirectional semantic consistency loss $\mathcal{L}_{\text{sem}}$ [48] to keep the bidirectional semantic consistency between the pseudoimages and corresponding original images. The bidirectional semantic consistency loss is expressed as follows:

$$\begin{aligned}\mathcal{L}_{\text{sem}}&(G_T, G_S, F_{\text{inter}}, F_S)\\ &= \sum_{h,w} KL\left(F_S(x_s)^{(h,w)} \| F_{inter}(G_T(x_s))^{(h,w)}\right)\\ &\quad + KL\left(F_{\text{inter}}(x_t)^{(h,w)} \| F_S(G_S(x_t))^{(h,w)}\right) \quad (7)\end{aligned}$$

where KL denotes the Kullback–Leibler divergence.

After the generator G_T is trained to produce satisfactory pseudotarget images $G_T(x_s)$, the inter-domain classifier F_{inter} could be trained on the pseudotarget images $G_T(x_s)$ and corresponding labels y_s. Therefore, we use target segmentation loss $\mathcal{L}_{\text{seg}}^{\text{inter}}$ to train and adapt the inter-domain classifier F_{inter} from source domain to target domain. The target segmentation loss $\mathcal{L}_{\text{seg}}^{\text{inter}}$ could be written as follows:

$$\begin{aligned}\mathcal{L}_{\text{seg}}^{\text{inter}}&(F_{\text{inter}}, G_T)\\ &= -\sum_{h,w}\sum_{c} y_s^{(h,w,c)} \log(F_{\text{inter}}(G_T(x_s)))^{(h,w,c)}. \quad (8)\end{aligned}$$

The target segmentation loss $\mathcal{L}_{\text{seg}}^{\text{inter}}$ is used to train and adapt the inter-domain classifier F_{inter} from source domain to target domain. Besides, to enhance the robustness of the UDA framework, we consider the inverse procedure and optimize the target-to-source direction of the InterDA module to form the bidirectional learning (BDL), so we apply source segmentation loss $\mathcal{L}_{\text{seg}}^{s}$ to train and adapt the source classifier F_S in the target-to-source direction. The source segmentation loss $\mathcal{L}_{\text{seg}}^{s}$ is defined as follows:

$$\mathcal{L}_{\text{seg}}^{s}(F_S) = -\sum_{h,w}\sum_{c} y_s^{(h,w,c)} \log(F_S(x_s))^{(h,w,c)}. \quad (9)$$

For the whole InterDA module, the loss function $\mathcal{L}_{\text{inter}}$ can be summarized as follows:

$$\begin{aligned}\mathcal{L}_{\text{inter}} &= \lambda_{\text{seg}}^{s}\mathcal{L}_{\text{seg}}^{s} + \lambda_{\text{seg}}^{\text{inter}}\mathcal{L}_{\text{seg}}^{\text{inter}} + \lambda_{\text{adv}}^{\text{inter}}\mathcal{L}_{\text{adv}}^{\text{inter}}\\ &\quad + \lambda_{\text{cyc}}\mathcal{L}_{\text{cyc}} + \lambda_{idt}\mathcal{L}_{\text{idt}} + \lambda_{\text{sem}}\mathcal{L}_{\text{sem}}. \quad (10)\end{aligned}$$

The goal of InterDA module is to learn a well-trained inter-domain segmentation model F_{inter} by optimizing the following equation:

$$F_{\text{inter}}^{*} = \min_{F_{\text{inter}}}\min_{F_S}\max_{G_T,G_S}\max_{D_S,D_T}\mathcal{L}_{\text{inter}}. \quad (11)$$

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2) Generator Selection Strategy: The key to the existing UDA methods based on I2I is the quality of image generators, which generate pseudoimages for the training of classifiers. The low-quality image generators may produce unsatisfactory pseudoimages, which would lead to poor performance of the inter-domain classifier trained on the pseudotarget images. Although the existing GAN-based I2I models usually work well in the UDA frameworks, they sometimes suffer from mode collapse and instability issues. These models directly use the generator saved after a fixed training epoch or the generator with the least training loss, and the generators selected by these ways would not be the best in many situations, which would significantly influence the performance of the inter-domain classifier. To handle the abovementioned issues, we propose a practical assessment approach for GAN-based I2I, which is called GSS, to evaluate all the intermediate generators quantitatively and choose the generator, which can generate best pseudoimages for the inter-domain segmentation model.

We assume that the number of whole training epochs for I2I network is M, and we cache the intermediate generators every m epochs. Thus, we can obtain $N = \lceil (M/m) \rceil$ pairs of intermediate models G_T^i and G_S^i ($i \in [1, N]$) when the whole training process is over. For each pair of intermediate generators, we compute the generator score S_i by the following equation:

$$S_i = \delta\big(F_S\big(G_T^i(x_s)\big), y_s\big) + \delta\big(F_S\big(G_S^i(x_s)\big), y_s\big) \tag{12}$$

where δ is a function, which computes the mean intersection over union (mIoU) value between the predicted results $F_S(G_T^i(x_s))$, $F_S(G_S^i(x_s))$ and the labels y_s. The first term $\delta(F_S(G_T^i(x_s)), y_s)$ is designed to assess the intermediate generator G_T^i, while the second term $\delta(F_S(G_S^i(x_s)), y_s)$ is designed to assess the intermediate generator G_S^i. We assume that the generator score S_i would be higher if the coupled intermediate models G_T^i and G_S^i are better at producing pseudoimages for training the classifier.

We rank all pairs of intermediate models according to their generator score S_i and select the coupled intermediate models G_T^i and G_S^i, which have the highest generator score, as the final I2I generators G_T and G_S.

B. ER Module

The ER module is designed to cluster all the target images into multiple subdomains on the basis of InterDA module. Although the domain gap between the source and target domains is effectively minimized after InterDA, the inter-domain classifier cannot output satisfactory predictions for all the target images due to the distribution discrepancy in the target domain. As illustrated in Fig. 1, a number of predicted results are precise and clean while a few predictions are bad and blurry. Based on this fact and built upon InterDA, we utilize the ER module to cluster all the target images into multiple subdomains according to the confidence level of the predicted results of the inter-domain classifier, so that we can treat the IntraDA problem in the target domain as the InterDA problem between subdomains. Because the target labels are unavailable, we evaluate the predicted results of the target images by computing the entropy maps, which are the mean entropy of softmax probability outputs P_t ($P_t = F_{\text{inter}}(x_t)$) of the inter-domain segmentation model [18]. The entropy maps represent the confidence levels of the predicted results, and they can be formulated as follows:

$$I_t^{(h,w)} = -\sum_c P_t^{(h,w,c)} \log P_t^{(h,w,c)}. \tag{13}$$

This is because we use the predicted results as the pseudolabels for the target images. The entropy maps also represent the confidence levels of the pseudolabels. Specifically, if the entropy is lower, the predicted results and pseudolabels would be more confident and accurate, and the effectiveness of pseudolabels would be better. Therefore, based on the entropy maps, we rank the predicted results with corresponding target images by computing the entropy score $R(x_t)$ [17] for each predicted map P_t and its corresponding target image x_t. The entropy score $R(x_t)$ is defined as follows:

$$R(x_t) = \frac{1}{HW} \sum_{h,w} I_t^{(h,w)}. \tag{14}$$

On this basis, IntraDANet [17] decomposed the target domain into two splits (clean split and noisy split); then, they used the clean split to align with the noisy split via self-supervision. However, in remote sensing, the target domain is usually complex and diverse, which means splitting the target domain into two subdomains is not enough to handle the distribution discrepancy within the target domain. In other words, there may still be some distribution discrepancies in the subdomains clustered by IntraDANet [17]. To solve the aforesaid issue, we propose to split the target domain into multiple (K) subdomains (subdomain $\mathcal{D}_1$, $\mathcal{D}_2$, ..., $\mathcal{D}_K$) and present an iterative IntraDA algorithm to iteratively reduce the complex and diverse intra-domain shift in field of remote sensing. It is worth noting that K is the hyperparameter, which is adopted to control the number of subdomains (iteration number) in the target domain, and we further study the influence of K in Section IV-E2.

C. Iterative Intra-Domain Adaptation Module

The existing InterDA methods are not enough to handle the complex intra-domain shift within the target domain in remote sensing. Thus, the iterative IntraDA module is designed to cope with this drawback by iteratively aligning the clean subdomain with multiple noisy subdomains on the basis of InterDA and ER. In this module, we align the subdomains in output space instead of image space, because the differences in image space are quite small within the target domain. Also, we apply SSL technology due to lack of annotations for the target images.

1) Pseudolabel Generation Strategy: Because the target labels are unavailable, it is infeasible to align the different subdomains directly. IntraDANet [17] proposed to employ the predicted maps of the target images as pseudolabels. Although the predicted results for the clean subdomain are low entropy and clean as a whole, there still exist a few high-entropy and wrong pixels in the clean images, which may mislead

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Algorithm 1 Iterative Intra-Domain Adaptation Algorithm

Input:
The target subdomains ($\mathcal{D}_1$, $\mathcal{D}_2$, ..., $\mathcal{D}_K$)
The clean subdomain $\mathcal{D}_{clean} = \emptyset$
The inter-domain classifier F_{inter}
The initial discriminator D_{intra}
Output:
The trained network $F_{intra}^{(K-1)}$, $D_{intra}^{(K-1)}$

$F_{intra}^{(0)} \leftarrow F_{inter}$
$D_{intra}^{(0)} \leftarrow D_{intra}$
for $i = 1$ to $K-1$ **do**
  input images $x_i \in \mathcal{D}_i$ into $F_{intra}^{(i-1)}$ and generate pseudolabels $\hat{y}_i$ for $x_i \in \mathcal{D}_i$ with Equation (15)
  add $(x_i, \hat{y}_i) \in \mathcal{D}_i$ into $\mathcal{D}_{clean}$
  use $(x_{clean}, \hat{y}_{clean}) \in \mathcal{D}_{clean}$ and $x_{i+1} \in \mathcal{D}_{i+1}$ to train $F_{intra}^{(i)} \leftarrow F_{intra}^{(i-1)}$, $D_{intra}^{(i)} \leftarrow D_{intra}^{(i-1)}$ with Equation (21)
end
return $F_{intra}^{(K-1)}$, $D_{intra}^{(K-1)}$

the segmentation model in the training process and degrade the performance of the segmentation model. To solve this problem, we propose a new PLGS, which takes advantages of the combination of softmax probability and entropy as the confidence assessment and removes both high-entropy and low-confident pixels in pseudolabels, to produce more reliable pseudolabels. Also, we formulate the PLGS for the trained segmentation network F_{inter} in the following equation:

$$\hat{y}_t^{(h,w,c)} = \begin{cases} c, & \text{if } P_t^{(h,w,c)} > \mu^c \text{ or } I_t^{(h,w)} < \upsilon^c \\ & \text{and } \operatorname{argmax}_{\tilde{c}} P_t^{(h,w,\tilde{c})} = c \\ 0, & \text{otherwise} \end{cases} \tag{15}$$

where μ^c is the probability threshold for class c, and υ^c is the entropy threshold for class c.

A simple approach [45] to determine the threshold μ^c can be expressed as follows:

$$\mu^c = \rho \left\{ P_t^{(h,w,c)} \mid x_t^{(h,w)} \wedge \operatorname*{argmax}_{\tilde{c}} P_t^{(h,w,\tilde{c})} = c \right\} \tag{16}$$

where $\wedge$ denotes logical conjunction, and the sorted softmax probability values $\{P_t^{(h,w,c)}\}$ correspond to pixels $\{x_t^{(h,w)}\}$, which belong to the input image x_t and have the highest probability of being category c. ρ is the hyperparameter (ρ is a percentage) to define the ratio between the preserved pixels and all pixels, and threshold μ^c is equal to the dividing value between the preserved pixels and removed pixels. Videlicet, we preserve the top ρ pixels whose softmax probability values are greater than μ^c and remove the bottom $1-\rho$ pixels whose softmax probability values are less than μ^c.

Similarly, we determine the threshold υ^c by the following equation:

$$\upsilon^c = \sigma \left\{ I_t^{(h,w)} \mid x_t^{(h,w)} \wedge \operatorname*{argmax}_{\tilde{c}} P_t^{(h,w,\tilde{c})} = c \right\} \tag{17}$$

where σ is the hyperparameter, such as ρ. This method ensures that we can preserve the top σ (σ is a percentage) of the predicted maps for each category based on entropy.

2) Iterative Intra-Domain Adaptation: Different from the existing one-stage IntraDA methods, which split the target domain into merely two subdomains and use the clean subdomain to adapt the noisy subdomain, we propose an iterative algorithm to better reduce the domain shift between the multiple subdomains ($\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_K$) by aligning the cleanest subdomain $\mathcal{D}_1$ with other subdomains ($\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_K$), as illustrated in Algorithm 1.

In each iteration, we apply the adversarial learning framework [18] to reduce the intra-domain gap between the clean subdomain $\mathcal{D}_i$ and the noisy subdomain $\mathcal{D}_j$. We train the intra-domain segmentation model F_{intra} to extract the domain-invariant features between different subdomains in the target domain and utilize F_{intra} to generate pseudolabels for the clean subdomain. The intra-domain classifier F_{intra} is optimized by minimizing the cross-entropy loss

$$\mathcal{L}_{\text{seg}}^{\text{intra}}(F_{\text{intra}}) = -\sum_{h,w}\sum_{c} \hat{y}_{ti}^{(h,w,c)} \log(F_{\text{intra}}(x_{ti}))^{(h,w,c)} \tag{18}$$

where image x_{ti} belongs to subdomain $\mathcal{D}_i$, and $\hat{y}_{ti}$ is the pseudolabels of x_{ti}.

Following the common adversarial learning methods, the intra-domain discriminator D_{intra} (see Fig. 4) learns to distinguish the predictions of x_{ti} ($x_{ti} \in \mathcal{D}_i$) from the predictions of x_{tj} ($x_{tj} \in \mathcal{D}_j$), while the segmentation model F_{intra} is trained to fool D_{intra}. The adversarial learning loss $\mathcal{L}_{\text{adv}}^{\text{intra}}$ for optimizing F_{intra} and D_{intra} can be formulated as follows:

$$\begin{aligned} \mathcal{L}_{\text{adv}}^{\text{intra}}(F_{\text{intra}}, D_{\text{intra}}) = &-\sum_{h,w} \log\left(1 - D_{\text{intra}}\left(I_{tj}^{(h,w)}\right)\right) \\ &+ \log\left(D_{\text{intra}}\left(I_{ti}^{(h,w)}\right)\right) \end{aligned} \tag{19}$$

where I_{ti} is the entropy map of $x_{ti} \in \mathcal{D}_i$, and I_{tj} is the entropy map of $x_{tj} \in \mathcal{D}_j$.

Furthermore, we minimize the entropy of predicted maps of x_{tj} by introducing the entropy loss [18] defined as follows:

$$\mathcal{L}_{\text{ent}}(F_{\text{intra}}) = \sum_{h,w} I_{tj}^{(h,w)}. \tag{20}$$

For each iteration in our iterative IntraDA module, the IntraDA loss $\mathcal{L}_{\text{intra}}$ can be summarized as follows:

$$\mathcal{L}_{\text{intra}} = \lambda_{\text{seg}}^{\text{intra}}\mathcal{L}_{\text{seg}}^{\text{intra}} + \lambda_{\text{adv}}^{\text{intra}}\mathcal{L}_{\text{adv}}^{\text{intra}} + \lambda_{\text{ent}}\mathcal{L}_{\text{ent}}. \tag{21}$$

In each iteration, our goal is to learn a intra-domain classifier F_{intra} by the following equation:

$$F_{\text{intra}}^* = \min_{F_{\text{intra}}} \max_{D_{\text{intra}}} \mathcal{L}_{\text{intra}}. \tag{22}$$

IV. Experiments and Analysis

A. Datasets Description

We conduct our experiments on the International Society for Photogrammetry and Remote Sensing (ISPRS) (WGII/4) 2D semantic segmentation benchmark datasets [50], which include open-source aerial images acquired from Potsdam city [51] and Vaihingen city [52]. The Potsdam dataset consists of 36 images (6000 × 6000 pixels) whose spatial resolution is 5 cm. The three channels of the Potsdam images we employed

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TABLE I
NUMBER OF PATCHES AND CLASS PERCENTAGES FOR DATASETS

City	Patches	Class percentages					
		Imp. surf.	Build.	Low veg.	Tree	Car	Back.
Potsdam	4598	29.8%	28.1%	21.0%	14.5%	1.8%	4.8%
Vaihingen	1414	29.3%	26.8%	19.3%	22.5%	1.4%	0.7%

correspond to the red, green, and blue channels. The Vaihingen dataset contains 33 images (about 2000 × 2000 pixels) with a spatial resolution of 9 cm. Different from the Potsdam images, the Vaihingen images comprise red, green, and infrared bands. Both datasets have semantic segmentation labels, which conclude buildings, low vegetations, trees, cars, and impervious surfaces. In this article, all the labeled Potsdam images are cropped into 512 × 512 patches without any overlap, while all the Vaihingen images and corresponding labels are cut into 512 × 512 patches with an overlap of 200 pixels. For the Potsdam dataset, we split all the labeled patches into two sets by randomly choosing 30% patches as test set and 70% patches as train set. Also, for the Vaihingen dataset, we build the train set and test set in the same way. In our experiments, we utilize the train set of one dataset as the source domain and the test set of the other dataset as the target domain. Table I shows the number of patches and the class percentages for the two datasets. Some sample patches of the Potsdam dataset and Vaihingen dataset are shown in Fig. 5.

1) Domain Gap Analysis: There are many differences between the Potsdam images and the Vaihingen images, such as spatial resolutions, structural representations of the classes, and spectral bands. First, the two datasets are provided with different spatial resolutions. As a result, the same object may be of different sizes in two datasets. Second, due to regional differences, the structural representations of the objects in the Potsdam dataset may be different from that of the same objects in the Vaihingen dataset. For example, the building styles and the roof colors of the two cities are very comparable. At last, the Potsdam images and the Vaihingen images comprise different kinds of spectral bands. Because of the influence of the infrared channel, the trees and low vegetations are red in the Vaihingen dataset, while the vegetations are green in the Potsdam dataset. Besides the aforementioned inter-domain gap, there exists the intra-domain gap in the same dataset. As shown in Fig. 5, the building styles and the roof colors of the buildings in the same dataset are complex and diverse.

B. Experimental Settings

1) Network Architecture: We utilize Deeplab-V2 [3] as the semantic segmentation network F. The generators of I2I network follow bidirectional domain adaptation network (BiFDANet) [48], as depicted in Fig. 3. They comprise four convolutional layers, nine residual blocks, three deconvolutional layers, and one convolutional layer to generate the pseudoimages and apply ReLU activation and instance normalization layer. For all the discriminators used in our experiments, we follow BiFDANet [48], which contains five convolutional layers and uses leaky ReLU (LReLU) activation function and instance normalization layer, as shown in Fig. 4.

2) Training Details: The proposed IterDANet is implemented in the PyTorch framework, and we can separately train the three modules in sequence. First, we train the InterDA module for 100 ($M = 100$) epochs by minimizing the loss function $\mathcal{L}_{\text{inter}}$ in (10), and we set the hyperparameters of λ_{seg}^{s}, $\lambda_{\text{seg}}^{\text{inter}}$, $\lambda_{\text{adv}}^{\text{inter}}$, λ_{cyc}, λ_{idt}, and λ_{sem} to 10, 10, 1, 10, 5, and 0, respectively. Besides, we set $m = 10$ and use the GSS to select the best generators G_T and G_S. We do not use the bidirectional semantic consistency loss $\mathcal{L}_{\text{sem}}$ in the first 100 epochs, because if the untrained segmentation model could not extract the semantic information well, $\mathcal{L}_{\text{sem}}$ may be detrimental to the training of generators and classifiers. After we obtain a well-trained target classifier, we set $\lambda_{\text{sem}} = 10$ to add $\mathcal{L}_{\text{sem}}$ and keep other parameters in (10) unchanged. Then, the InterDA module is optimized for 100 epochs. For the first step, the generators, the discriminators, and the classifiers are jointly trained with the Adam optimizer [53]. We set batch size to 12 and set learning rate to 10^{-4}. Second, we apply the trained target classifier F_{inter} to segment all the target images and compute the score $R(x_t)$ (14) of the predictions. Then, we rank the target images based on the score $R(x_t)$ and split them evenly into four subdomains ($K = 4$). Third, we train the IntraDA module with Algorithm 1. For generating good-quality pseudolabels, we set $\rho = 75$ in (16) to compute softmax probability threshold μ^c for each class c and set $\sigma = 70$ in (17) to compute entropy threshold υ^c for each class c. In each iteration of training, we adopt the Adam optimizer [53] with $\beta_1 = 0.9$ and $\beta_2 = 0.999$, and the learning rates are set to 10^{-4} for the intra-domain classifier F_{intra} and discriminator D_{intra}. We set $\lambda_{\text{seg}}^{\text{intra}} = 1$, $\lambda_{adv}^{\text{intra}} = 0.01$, and $\lambda_{\text{ent}} = 0.001$ in (21), and we jointly optimize F_{intra} and D_{intra} for 50 epochs and update the clean subdomain and the target subdomain in each iteration. It is notable that the networks of the InterDA module would not be trained during the training process of the IntraDA module, and for all the hyperparameters in IterDANet, the values are obtained through empirical trials.

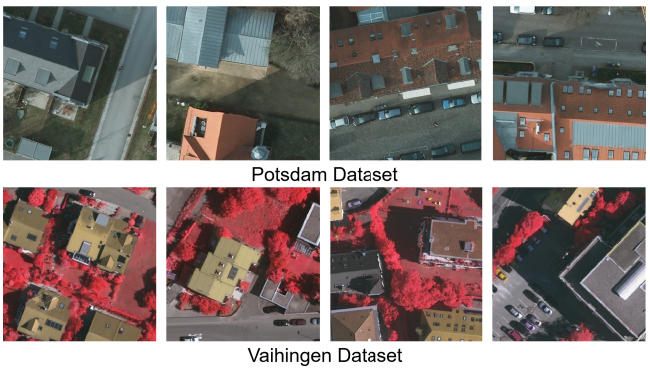

Fig. 5. Images sampled from the Potsdam and Vaihingen datasets.

3) Evaluation Metrics: In this article, we utilize *F1*-score ($F1$) and IoU [54] to evaluate all the approaches quantitatively, and these two metrics are defined as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{23}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{24}$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{25}$$

$$\text{IoU} = \frac{TP}{TP + FN + FP} \tag{26}$$

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TABLE II

COMPARING RESULTS (%) WITH THE COMPARISON METHODS ON THE ADAPTATION FROM POTSDAM TO VAIHINGEN. IMP SURF CORRESPONDS TO IMPERVIOUS SURFACES, AND LOW VEG CORRESPONDS TO LOW VEGETATION

Source: Potsdam, Target: Vaihingen														
Method	Imp surf		Building		Low veg		Tree		Car		Background		Avg	
	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	mF1	mIoU
Deeplab-V2(baseline)	33.8	20.4	47.5	33.5	21.2	12.7	0.0	0.0	17.1	10.2	8.0	4.8	21.3	13.6
Histogram matching	58.3	45.5	60.6	49.0	44.5	27.3	16.5	9.0	9.9	5.2	5.2	2.7	32.5	23.1
CycleGAN	63.4	46.4	62.4	57.6	61.0	39.8	55.0	37.6	20.9	11.7	16.7	9.1	46.5	33.7
AdaptSegNet(s)	67.3	51.1	61.8	57.1	39.8	26.4	39.2	21.8	24.5	14.9	12.6	7.2	41.0	29.7
AdaptSegNet(m)	69.7	53.6	65.2	62.5	57.4	36.0	43.8	26.4	31.8	20.0	17.5	10.4	47.6	34.8
MinEnt	70.7	53.6	63.8	60.7	44.9	29.7	42.1	23.4	28.1	16.0	19.4	9.2	44.8	32.1
AdvEnt	70.8	54.6	69.1	64.0	58.8	35.7	50.3	33.2	27.8	17.6	20.1	11.4	49.5	36.1
CycleGAN+SSL	65.0	48.4	63.1	60.0	67.7	46.8	53.1	35.7	31.7	19.6	17.1	8.5	49.6	36.5
BDL	68.6	52.3	62.2	59.6	68.6	47.9	62.4	43.7	30.4	17.0	21.8	11.5	52.3	38.7
FSDAN	67.7	50.5	69.3	63.5	65.8	43.8	63.4	46.8	29.6	17.1	**23.0**	**13.8**	53.1	39.3
IntraDANet	67.8	51.4	64.8	62.1	65.6	43.5	58.1	39.1	34.4	20.3	22.8	13.1	52.2	38.3
IterDANet(ours)	**71.5**	**55.7**	**70.6**	**65.5**	**70.9**	**51.3**	**66.7**	**49.6**	**35.1**	**20.5**	18.8	10.5	**55.6**	**42.2**
Supervised method	84.2	72.7	87.2	75.5	84.7	73.7	84.3	72.9	68.7	52.4	56.4	45.2	77.6	65.4

where FP, FN, and TP correspond to false positive, false negative, and true positive, respectively.

C. Comparing Methods

1) Deeplab-V2 [3]: Without utilizing any UDA approaches, we train a Deeplab-V2 on the source domain and directly classify the target images.

2) Histogram Matching [55]: The histogram of the source domain is matched with that of the target domain for each band. Then, the matched domain is used for training the target segmentation model.

3) CycleGAN [11]: This approach utilizes two GANs to perform the source-to-target and target-to-source translation while keeping the cycle consistency during the I2I. It aims at aligning the data distributions of the source domain and target domain in the image space. We use CycleGAN to translate the source images to the target domain and train the target classifier on the translated domain.

4) AdaptSegNet [15]: This adversarial learning approach trains a domain invariant segmentation model to predict labels for the source and target domain and trains the discriminator to align the source domain and target domain in the output space. AdaptSegNet(m) performs multilevel adversarial learning, which applies two discriminators to align the output of the last two classification layers. AdaptSegNet(s) uses single-level adversarial learning and aligns the output of the last one classification layer by one discriminator.

5) MinEnt [18]: This adversarial learning method trains a domain invariant classifier to generate predictions for the source and target images and simply minimizes the entropy of predictions without using discriminators.

6) AdvEnt [18]: This adversarial learning approach trains a domain invariant classifier to classify the source and target images and trains the discriminator to align the entropy maps of the predictions of the source and target images.

7) CycleGAN + SSL [43]: This method first performs InterDA based on CycleGAN [11], and then, it utilizes the inter-domain classifier to generate pseudolabels for the target domain. Finally, it optimizes the target classifier with the SSL method used in [43].

8) BDL [16]: BDL proposes to alternatively train the segmentation adaptation network and the I2I translation network in a bidirectional manner; then, it optimizes the segmentation adaptation network by SSL.

9) FSDAN [9]: This method first aligns the source and target domains in both image space and feature space by I2I and adversarial learning and then aligns the source and target domains in the output space by adversarial learning.

10) IntraDANet [17]: This approach first adapts the inter-domain classifier with adversarial learning [18]. Then, it splits the target domain into clean and noisy subdomains based on the entropy of the predictions. Finally, it optimizes the target intra-domain classifier from the clean subdomain to the noisy subdomain with adversarial learning [18].

For the abovementioned comparing models, we apply the same network architectures and training parameters as our proposed IterDANet for a fair comparison.

D. Results and Discussion

To validate the effectiveness and competitiveness of the proposed framework, we conduct extensive experiments on the Potsdam dataset and the Vaihingen dataset. For all the comparing methods and our IterDANet, Tables II and III report $F1$-score and IoU values for each class and mean $F1$-score and mean IoU value on the test set. Table IV depicts the performance gap between the UDA methods and the supervised method. The Deeplab-V2 and supervised method rows refer to the baseline no-adaptation method and ceiling fully supervised approach. The Histogram matching and CycleGAN rows correspond to UDA methods based on the I2I. The AdaptSegNet(s/m), MinEnt, and AdvEnt rows correspond to the UDA approaches based on adversarial learning. The CycleGAN + SSL, BDL, full-space domain adaptation network (FSDAN), IntraDANet, and IterDANet rows contain the UDA methods, which further optimize the classifier on the basis of InterDA.

Table II depicts the comparison results of the adaptation from Potsdam to Vaihingen. Because of the huge domain shift between the Potsdam and Vaihingen datasets, the baseline

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TABLE III

COMPARING RESULTS (%) WITH THE COMPARISON METHODS ON THE ADAPTATION FROM VAIHINGEN TO POTSDAM. IMP SURF CORRESPONDS TO IMPERVIOUS SURFACES, AND LOW VEG CORRESPONDS TO LOW VEGETATION

Source: Vaihingen, Target: Potsdam														
Method	Imp surf		Building		Low veg		Tree		Car		Background		Avg	
	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	mF1	mIoU
Deeplab-V2(baseline)	45.3	29.3	28.5	16.4	3.9	2.1	5.4	2.8	20.6	12.2	0.7	0.3	17.4	10.5
Histogram matching	62.6	45.6	38.8	24.0	57.8	40.7	19.2	10.6	10.2	5.4	0.0	0.0	31.4	21.1
CycleGAN	65.9	51.5	71.0	55.0	54.1	37.1	43.4	27.7	37.7	20.6	6.5	4.0	46.5	32.7
AdaptSegNet(s)	70.2	55.1	54.4	37.4	35.5	21.6	12.8	6.9	43.3	26.3	14.7	8.6	38.5	26.0
AdaptSegNet(m)	65.7	50.9	65.3	48.4	41.3	26.9	40.9	25.8	44.1	27.5	**19.9**	**11.5**	46.2	31.8
MinEnt	64.6	49.8	63.6	46.6	51.4	34.6	46.9	30.6	28.9	12.4	6.2	4.1	43.6	29.7
AdvEnt	65.5	50.7	68.4	52.0	51.5	34.7	47.1	30.8	**45.3**	**28.5**	10.9	6.5	48.1	33.9
CycleGAN+SSL	71.1	57.0	75.7	58.9	53.1	38.1	54.9	36.8	36.5	19.4	7.4	4.7	49.8	35.8
BDL	72.5	57.4	76.1	60.9	57.7	40.6	52.9	37.2	41.7	25.4	6.7	5.3	51.3	37.8
FSDAN	73.6	58.1	70.8	61.7	59.8	41.1	52.4	36.2	44.1	26.2	12.8	6.4	52.2	38.3
IntraDANet	73.8	58.6	71.9	56.2	58.0	40.9	50.5	33.8	44.3	27.4	13.2	7.7	51.9	37.4
IterDANet(ours)	**75.4**	**59.3**	**76.9**	**62.4**	**65.8**	**51.8**	**61.7**	**42.6**	39.4	24.3	6.1	4.2	**54.2**	**40.8**
Supervised method	86.0	76.5	88.2	78.3	85.3	74.1	81.3	68.1	74.3	58.9	72.2	55.9	81.2	68.6

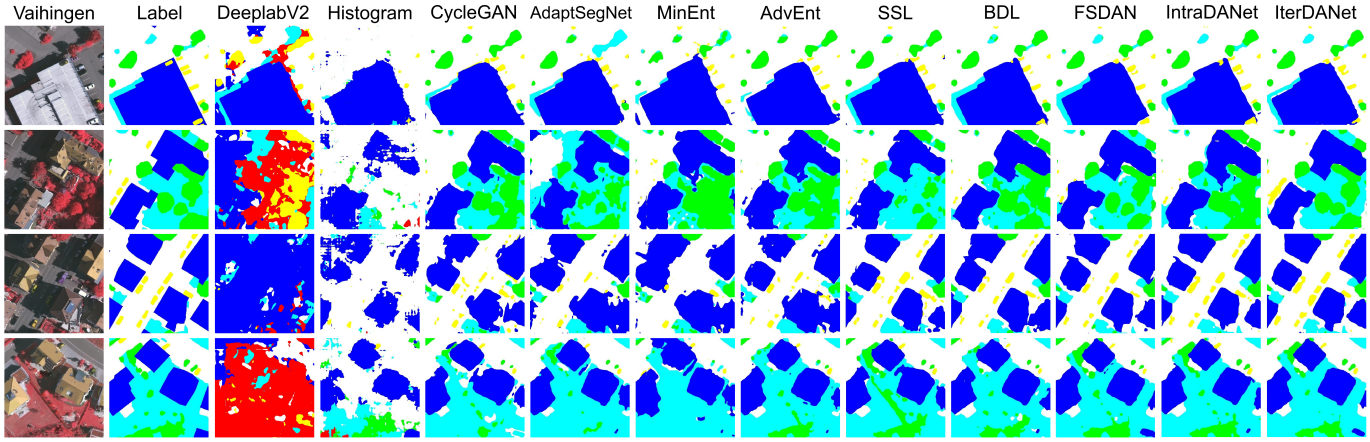

Fig. 6. Visual segmentation results of all the comparison methods on the adaptation from Potsdam domain to Vaihingen domain. Histogram and SSL denote Histogram matching and CycleGAN + SSL. White, blue, cyan, green, yellow, and red colors represent impervious surfaces, buildings, low vegetation, trees, cars, and background, respectively.

method Deeplab-V2, which does not apply any UDA methods, performs the worst of all comparison methods. The m$F1$ is 21.3%, and the mIoU is only 13.6%. All the UDA approaches improve the segmentation performance more or less, and the simplest Histogram matching achieves 11.2% m$F1$ and 9.5% mIoU improvement, which shows the effectiveness of the UDA approaches. Among all the unsupervised methods, our proposed IterDANet has the highest m$F1$ score and mIoU value, and it outperforms the runner-up IntraDANet by 3.4% for m$F1$ and 3.9% for mIoU. Furthermore, compared with other UDA approaches, the proposed IterDANet effectively narrows the performance gap with the supervised method.

Transferring from Vaihingen to Potsdam is more difficult than transferring from Potsdam to Vaihingen, because the number of patches of Potsdam is much more than that of Vaihingen, and the Potsdam dataset is more complex and more diverse than the Vaihingen dataset. To validate the robustness of the proposed IterDANet, we perform experiments and report classification results in Tables III and IV. The baseline approach Deeplab-V2 still achieves the lowest m$F1$ score and mIoU value, because there exists a large distribution shift. With the help of InterDA and iterative IntraDA, our IterDANet still performs better than other comparing approaches, and it achieves 54.2% m$F1$ and 40.8% mIoU, which improves the baseline by 36.8% for m$F1$ and 30.3% for mIoU. Table IV demonstrates that our proposed IterDANet can further reduce the performance gap with supervised methods, even if the adaptation is more difficult.

Figs. 6 and 7 show the segmentation results for Deeplab-V2, Histogram matching (Histogram), CycleGAN, AdaptSegNet(m), MinEnt, AdvEnt, CycleGAN + SSL (SSL), IntraDANet, and IterDANet. Without UDA techniques, the baseline Deeplab-V2 outputs poor and chaotic predictions. The simplest UDA method Histogram has a better performance than Deeplab-V2, which shows the importance of UDA. Among all the UDA models, our proposed IterDANet, which performs InterDA and minimizes the intra-domain gap iteratively, achieves the most accurate and cleanest classification predictions for almost all classes.

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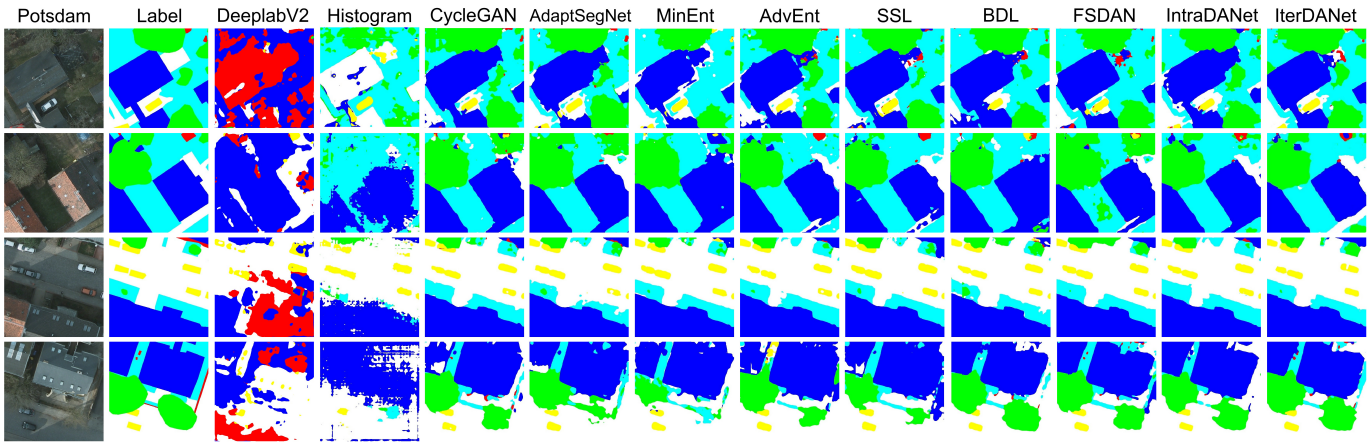

Fig. 7. Visual segmentation results of all the comparison methods on the adaptation from Vaihingen domain to Potsdam domain. Histogram and SSL denote Histogram matching and CycleGAN + SSL. White, blue, cyan, green, yellow, and red colors represent impervious surfaces, buildings, low vegetation, trees, cars, and background, respectively.

TABLE IV
PERFORMANCE GAP [mIoU(%)] BETWEEN THE UDA METHODS AND THE SUPERVISED METHOD (ORACLE) ON THE POTSDAM AND VAIHINGEN DATASETS

	Potsdam → Vaihingen			Vaihingen → Potsdam		
Method	UDA	Oracle	Gap	UDA	Oracle	Gap
Deeplab-V2(baseline)	13.6	65.4	-51.8	10.5	68.6	-58.1
Histogram matching	23.1	65.4	-42.3	21.1	68.6	-47.5
CycleGAN	33.7	65.4	-31.7	32.7	68.6	-35.9
AdaptSegNet(s)	29.7	65.4	-35.7	26.0	68.6	-42.6
AdaptSegNet(m)	34.8	65.4	-30.6	31.8	68.6	-36.8
MinEnt	32.1	65.4	-33.3	29.7	68.6	-38.9
AdvEnt	36.1	65.4	-29.3	33.9	68.6	-34.7
CycleGAN+SSL	36.5	65.4	-28.9	35.8	68.6	-32.8
BDL	38.7	65.4	-26.7	37.8	68.6	-30.8
FSDAN	39.3	65.4	-26.1	38.3	68.6	-30.3
IntraDANet	38.3	65.4	-27.1	37.4	68.6	-31.2
IterDANet(ours)	**42.2**	65.4	**-23.2**	**40.8**	68.6	**-27.8**

1) IterDANet Versus Image Translation: As illustrated in Tables II and III, Histogram matching performs the worst of all the UDA methods. This is because Histogram matching only matches the histograms of the source and target domain, which ignores the distributions in the image space. In comparison, when generating pseudotarget images, IterDANet and CycleGAN utilize a discriminator to determine whether the images are pseudotarget images generated by the generator or original target images. Thus, the generator must imitate the high-level distributions of the target domain to fool the discriminator. In other words, the pseudotarget images generated by IterDANet and CycleGAN are more similar to the original target images in the image space. So, the proposed IterDANet and CycleGAN have better performances than Histogram matching.

Although CycleGAN leans the high-lever distributions in the image space, it could not always maintain the semantic contents when performing I2I. For example, the buildings may be turned into trees during I2I. Another problem is that the image quality of CycleGAN is unstable. There may exist some grid-like noises and artificiality in the pseudoimages generated by CycleGAN. These factors lead to mismatches between generated images and labels, which would degrade the performance of the classifier. By contrast, our IterDANet adopts the bidirectional semantic consistency constraint to keep the semantic contents, and we propose the GSS, which choose the better generator to produce high-quality pseudoimages for the training. Therefore, the proposed IterDANet outperforms CycleGAN by about 8% for mIoU as reported in Tables II and III.

2) IterDANet Versus Adversarial Learning: AdaptSegNet assumes that the contexts and layouts of the source and target domain are similar, and it aligns the output of the segmentation model, which takes the source images and target images as input. If AdaptSegNet aligns the output of the classifier in multiple layers, it could achieve better performance. In the Potsdam and Vaihingen datasets, there exists large domain shift in the classes of low vegetation and tree, while the domain shift between the impervious surfaces and building is smaller. Due to the influence of the infrared band, the green low vegetation and trees in the Potsdam dataset turn red in the Vaihingen dataset, which makes it difficult for the classifier trained on the Potsdam dataset to segment the low vegetation and trees in the vegetation images. Thus, the performance of AdaptSegNet is poor in the classes of low vegetation and trees, as shown in Tables II and III. When performing InterDA, our proposed IterDANet transforms the color and distributions of low vegetation and trees from the Potsdam domain to the Vaihingen domain. The inter-domain classifier could learn the features of trees and low vegetation of the target domain better. Therefore, IterDANet obviously outperforms AdaptSegNet on the $F1$ and IoU of the low vegetation and trees. On the other hand, although AdaptSegNet works well in the classes of impervious surfaces and building, it does not consider the intra-domain gap in the target domain. As a result, the classifier still outputs some unsatisfactory predictions in some cases. In consideration of the intra-domain gap, our IterDANet further performs IntraDA within the target domain and optimize the classifier to output better predicted results of the images, which are hard to segment. With the help of the IntraDA, our proposed IterDANet surpasses AdaptSegNet on the m$F1$ score and mIoU value. AdvEnt and MinEnt

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enforce the segmentation model to generate low-entropy and confident predicted maps based on the adversarial loss and entropy loss. Even though AdvEnt and MinEnt further improve the performance of adversarial learning methods in some situations, they still suffer from the spectral variations and intra-domain gap problems as the existing adversarial learning methods. So AdvEnt and MinEnt perform worse than our proposed IterDANet.

3) IterDANet Versus SSL: BDL and CycleGAN + SSL perform InterDA on the basis of GAN-based I2I (e.g., CycleGAN) and further optimize the target segmentation model via SSL; therefore, they have better performances than the InterDA methods, as illustrated in Tables II and III. However, because of the intra-domain gap, there are still some high-entropy and low-confident predictions in the pseudolabels, such wrong information may mislead the classifier in the self-training process, and the performance of the classifier would be highly influenced. In contrast, IterDANet iteratively optimizes the low-confident and high-entropy pseudolabels and makes them more accurate. Therefore, IterDANet achieves higher m$F1$ score and mIoU values than BDL and CycleGAN + SSL.

4) IterDANet Versus IntraDA: When performing IntraDA, IntraDANet splits the target domain into two subdomains based on the entropy of the predictions; then, it employs the low-entropy subdomain to adapt the high-entropy subdomain. However, it does not take into account the complexity and diversity of the target domain in remote sensing, which cannot effectively reduce the intra-domain gap. Besides, it directly utilizes the predictions of the inter-domain classifier as the pseudolabels for the target images, and the incorrect pixels in the pseudolabels may mislead the target segmentation model in the IntraDA process. For IterDANet, it divides the target domain into multisubdomain to deal with the diverse distributions in remote sensing images. Furthermore, the proposed PLGS removes the high-entropy and low-confident pixels in the pseudolabels, which makes the pseudolabels more accurate. Therefore, our IterDANet has a better performance than IntraDANet.

5) IterDANet Versus FSDAN: To reduce the inter-domain shift, FSDAN first aligns the source and target domains in the image and feature space; then, it aligns the output space. As shown in Tables II and III, FSDAN achieves the second best performance among all the comparing approaches. However, due to the limitations of intra-domain shift in the target domain, the mean IoU performance of FSDAN is inferior to that of IterDANet.

E. Model Analysis

To better evaluate and analyze our proposed framework, extensive studies are conducted on the Potsdam and Vaihingen datasets. First, we investigate the effect of each component in our IterDANet. Second, we discuss the influence of the iteration number (number of subdomains) K. Third, we conduct the computational analysis for the IterDANet. Finally, we validate the effectiveness of the GSS and PLGS.

1) Components Analysis: As we have introduced in Section III, the core ideas of the proposed framework are as follows: 1) the proposed iterative IntraDA algorithm to cluster the target domain into multiple subdomains and reduce the intra-domain shift iteratively; 2) the proposed GSS to access and choose a well-trained generator to generate pseudoimages without artifacts and noises for the training of inter-domain classifier; and 3) the proposed PLGS to produce more accurate pseudolabels for the SSL in the iterative IntraDA module. To validate the role of abovementioned three components in IterDANet, we take the Potsdam dataset as the labeled source domain and the Vaihingen dataset as the unlabeled target domain and perform ablation experiments. Table V reports the quantitative results of utilizing different component parts. We utilize the bidirectional domain adaptation method (i.e., BiFDANet [48]) as the baseline method for the InterDA module and IntraDA [17] as the baseline method for the IntraDA module.

TABLE V

ABLATION ANALYSIS OF EACH COMPONENT IN OUR FRAMEWORK ON THE POTSDAM AND VAIHINGEN DATASETS. INTER-DA DENOTES THE BASELINE INTERDA METHOD, AND INTRA-DA IS THE BASELINE INTRADA METHOD. MTL IS THE EXISTING GSS. GSS DENOTES THE PROPOSED GSS, AND PLGS IS THE PROPOSED PSEUDOLABEL GENERATION STRATEGY. ESL IS THE EXISTING ENTROPY-BASED PLGS, WHILE SSL IS THE EXISTING SOFTMAX PROBABILITY-BASED PLGS. ITER-DA DENOTES THE PROPOSED ITERATIVE INTRADA ALGORITHM

Source: Potsdam, Target: Vaihingen		
Method	Components	mIoU(%)
Model-I	Inter-DA	33.6
Model-II	Inter-DA+MTL	35.4
Model-III	Inter-DA+GSS	36.3
Model-IV	Inter-DA+GSS+Intra-DA	38.5
Model-V	Inter-DA+GSS+Intra-DA+ESL	39.1
Model-VI	Inter-DA+GSS+Intra-DA+SSL	38.8
Model-VII	Inter-DA+GSS+Intra-DA+PLGS	39.6
Model-VIII	Inter-DA+GSS+Iter-DA+PLGS	**42.2**

As shown in Table V, Intra-DA corresponds to the baseline IntraDA method. Inter-DA denotes the baseline InterDA method, which directly uses the image generator saved after the final epoch. MTL denotes using the generator with minimum training loss, while GSS corresponds to the proposed GSS. SSL is the existing softmax probability-based PLGS. Entropy-based self-supervised learning (ESL) denotes the existing entropy-based PLGS, while PLGS denotes our proposed pseudo label generation strategy. Finally, Iter-DA corresponds to the proposed iterative IntraDA algorithm. Compared with Model-I and Model-II, Model-III has a better performance with the help of the proposed GSS, which highlights the effectiveness of the proposed GSS. Then, from the result of Model-IV, we can see that the baseline IntraDA method (Intra-DA) leads to further performance improvement, which verifies the superiority and necessity of the IntraDA methods. Besides, Model-V and Model-VI perform better than Model-IV, because there are some incorrect pixels in the predictions of the inter-domain classifier, and the existing ESL and SSL could remove a few incorrect pixels in the pseudolabels. Compared with Model-V and Model-VI, Model-VII with our proposed PLGS achieves better performance, which shows the effectiveness and superiority of our proposed PLGS against the existing

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TABLE VI

INFLUENCE OF ITERATION NUMBER (NUMBER OF SUBDOMAINS) K FOR INTRA-DOMAIN ADAPTATION ON THE POTSDAM AND VAIHINGEN DATASETS

Source: Potsdam, Target: Vaihingen						
K	1	2	3	4	6	8
mIoU (%)	38.5	39.6	40.8	**42.2**	41.8	41.1

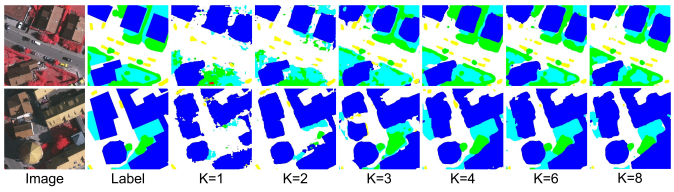

Fig. 8. Visual segmentation results of IterDANet with different iteration numbers (number of subdomains) K in the Vaihingen → Potsdam experiment. White, blue, cyan, green, yellow, and red colors represent impervious surfaces, buildings, low vegetation, trees, cars, and background, respectively.

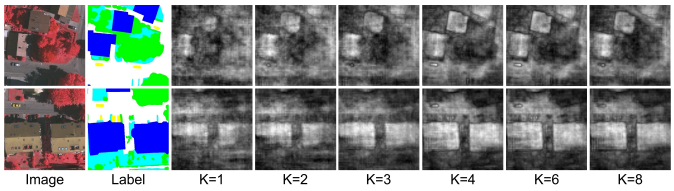

Fig. 9. Entropy maps of IterDANet with different iteration numbers (number of subdomains) K in the Vaihingen → Potsdam experiment.

entropy-based and softmax probability-based PLGS. At last, Model-VIII using the proposed iterative IntraDA algorithm (Iter-DA) performs the best among all the models due to the superiority of the proposed iterative IntraDA algorithm against the existing IntraDA methods.

2) Influence of Iteration Number K: The iteration number (number of subdomains) K for IntraDA module is an important hyperparameter for our method. To study the influence of iteration number (number of subdomains) K for optimization, we repeatedly train our framework with different iteration numbers K in Algorithm 1. The segmentation results (mIoU) are reported in Table VI. Also, Figs. 8 and 9 show the visual segmentation results and entropy maps of our IterDANet with different iteration numbers K.

In some way, when the iteration number (number of subdomains) is one ($K = 1$), we can see the framework as the improved SSL method. Also, when the iteration number (number of subdomains) is two ($K = 2$), we can regard the framework as the improved IntraDANet. We can find that the mean IoU value is improved, as the number of iterations increases. When the number of iterations (number of subdomains) reaches four ($K = 4$), the framework achieves the highest mIoU value. Then, when the number of iterations continues to increase, the performance of the framework slightly degrades but is still better than the improved SSL method ($K = 1$) and IntraDANet ($K = 2$). In Figs. 8 and 9, along with the increasing of iteration numbers, the predictions of the model are becoming more and more accurate, and the entropy of the predictions gets lower and lower until the iteration number reaches 4; then, the performance of the model gets slightly worse. It means that the framework does not always perform better with more iterations. One possible reason is that too many iteration numbers and target subdomains lead to lack of training sample at each iteration, which influences the performance of our framework.

TABLE VII

SEGMENTATION RESULTS (%), TOTAL TRAINING TIME, AND TEST TIME FOR EVERY PATCH OF THE COMPARISON METHODS ON THE POTSDAM AND VAIHINGEN DATASETS

Source: Potsdam, Target: Vaihingen			
Method	mIoU(%)	Training time (h:m:s)	Test time (s)
Histogram matching	23.1	**4:13:14**	0.26
CycleGAN	33.7	17:53:20	0.26
AdaptSegNet	34.8	8:12:51	0.26
AdvEnt	36.1	8:14:38	0.26
CycleGAN+SSL	36.5	22:07:43	0.26
BDL	38.7	24:57:24	0.26
FSDAN	39.3	31:16:33	0.26
IntraDANet	38.3	18:05:43	0.26
IterDANet	**42.2**	30:46:43	0.26

3) Computational Analysis: To comprehensively compare the evaluation indicators and running time of various methods, we perform experiments on the adaptation from Potsdam domain and Vaihingen domain. All the comparing methods are trained on the single NVIDIA Tesla P40 GPU with the same training parameters and networks, and we report the total training time and test time per patch for different methods in Table VII. As shown in Table VII, all the methods take the same test time, because all of them directly use the classifier with the same architecture to segment the same target images at test stage. It is worth noting that the IterDANet does not cost any additional computation during the testing phase. When it comes to the training time, the Histogram matching takes the least amount of training time due to its simplicity, but this method performs the worst among all the methods. For the InterDA methods, although CycleGAN takes more training time than AdvEnt and AdaptSegNet, because training the generators and discriminators of the I2I model costs lots of training time, the I2I translation model leads to performance improvements in the classes of low vegetation and tree, which are significantly influenced by the differences in spectral bands. It is notable that the two-stage CycleGAN + SSL, BDL, FSDAN, IntraDANet, and IterDANet take much more training time than the one-stage InterDA methods (AdaptSegNet, AdvEnt, and Histogram matching), because these two-stage methods further train the inter-domain classifier to perform SSL or IntraDA, which further improves the performance of the inter-domain classifier. Also, among these two-stage methods, our IterDANet takes the second most training time but achieves the best performance. For the memory cost, the memory cost of InterDA module of IterDANet is about the same as the common UDA frameworks based on I2I, while the memory cost of the IntraDA module of IterDANet is about the same as the common UDA frameworks based on adversarial learning. Because our IterDANet can train the InterDA module and IntraDA module separately, the memory cost of IterDANet corresponds to the maximum amount of memory consumption of the two modules instead

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TABLE VIII

GENERATOR SCORES (%) AND SEGMENTATION RESULTS (%) OF INTERMEDIATE GENERATORS SAVED AFTER DIFFERENT NUMBERS OF TRAINING EPOCHS ON THE POTSDAM AND VAIHINGEN DATASETS

Source: Potsdam, Target: Vaihingen										
Epoch	10	20	30	40	50	60	70	80	90	100
Score	66.4	85.2	99.8	98.5	102.1	**103.8**	103.2	103.5	103.4	101.6
mIoU	11.8	14.8	31.6	30.1	34.9	**36.3**	35.5	35.4	35.8	33.6

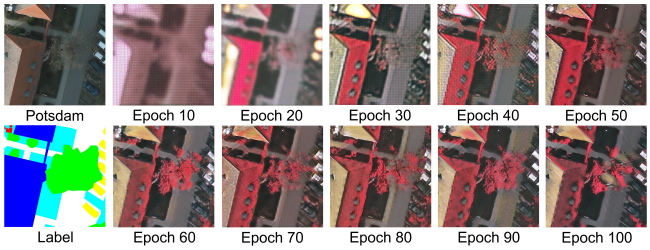

Fig. 10. Pseudotarget images generated by intermediate generators saved after different numbers of training epochs.

TABLE IX

SEGMENTATION RESULTS (%) OF IterDANet WITH DIFFERENT PSEUDOLABEL GENERATION STRATEGIES ON THE POTSDAM AND VAIHINGEN DATASETS

Source: Potsdam, Target: Vaihingen				
method	No	SSL	ESL	Ours
mIoU	36.7	37.9	38.1	**38.4**

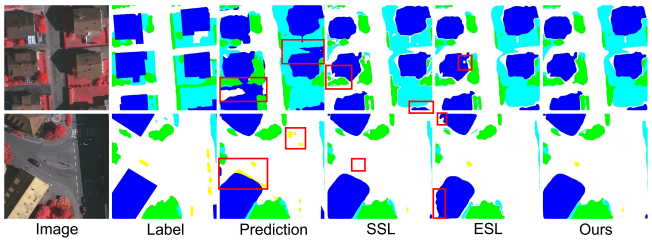

Fig. 11. Pseudolabels generated by different pseudolabel generation strategies. SSL and ESL denote softmax probability-based and entropy-based SSL methods. White, blue, cyan, green, yellow, and red colors represent impervious surfaces, buildings, low vegetation, trees, cars, and background, respectively.

of the sum of memory consumption. Therefore, the proposed IterDANet would not consume much more memory than the existing UDA methods.

4) Analysis of GSS: To verify the effectiveness of our proposed GSS, we perform experiments on the adaptation from Potsdam domain and Vaihingen domain. As mentioned in Section III-A, we save the intermediate generators every ten epochs during the training process and apply them to generate pseudotarget images, as shown in Fig. 10. Then, the target classifiers are trained on the pseudotarget images generated by different intermediate generators and their corresponding labels. Table VIII depicts the generator scores and segmentation results (mIoU) of the different intermediate generators.

From Fig. 10, we can see that the performance of generator is getting better and better in the first 30 training epochs. At first glance, it seems that all the intermediate generators saved after 30 training epochs work well, and we cannot directly pick out the best pseudotarget images with human eyes. The intermediate generator saved after more training epochs does not perform better the intermediate generator saved after less training epochs. In another word, the image quality does not depend on the training epochs of generators. Also, there are obvious artifacts on the trees in the images generated by the last intermediate generators (epoch $= 100$). It is because the GAN-based models suffer from instability issues. To deal with this problem, we propose the GSS to quantitatively evaluate each intermediate generator. As shown in Table VIII, the intermediate generator with the highest generator score has the best performance among all the intermediate generators and leads to 2.7% mIoU improvement compared with the final intermediate generator, which validates the effectiveness of the proposed GSS.

5) Analysis of PLGS: To further analyze the proposed PLGS, we conduct a series of experiments on the Potsdam and Vaihingen datasets. First, we compare our proposed PLGS with the existing PLGS used in the most of SSL frameworks. Then, we describe how to select the values of hyperparameters ρ and σ and investigate the influence of these two hyperparameters.

Table IX reports the comparison results (mIoU) among no PLGS (No), softmax probability-based strategy (SSL) [16], ESL [45], and our proposed approach (Ours). It is worth noting that all the models with pseudolabel generation strategies have better performances than the model without PLGS, and our proposed strategy yields the highest mIoU values among all the strategies, which shows the superiority of the proposed PLGS. Fig. 11 shows the pseudolabels generated by different approaches. We can see that without PLGS, there are many incorrect classification results in the predictions of the target classifier as highlighted by red rectangle in the third column (prediction). Although SSL [16] and ESL [45] filter out a few incorrect pixels of the predictions, they move out some correct and useful pixels at the same time as indicated by red rectangle in the fourth and fifth columns. Compared with SSL [16] and ESL [45], our strategy removes both high-entropy and low-confident pixels and preserves the low-entropy and high-confident pixels as many as possible, which leads to a better performance.

When choosing the hyperparameters ρ and σ in (17) and (16), we need to balance the quantity and quality of the pseudolabels effectively. If ρ and σ are too small, the preserved pseudolabels would be sufficiently low entropy and high confident, but there will not be enough pseudolabels for training the target segmentation model. On the flip side, if ρ and σ are too large, although there are sufficient pseudolabels for training, the incorrect predictions cannot be removed as many as possible. In other words, there still exist a few high-entropy and low-confident pixels in the predictions, which would mislead the classifier in the self-training process and result in the poor performance of the target classifier. To deal with these problems, we show the relationship between pixel ratio ρ (ratio of preserved pseudolabels to all predictions) and prediction confidence (softmax probability of each pixels in the predictions) in Fig. 12 (left). Then, we depict the relationship between pixel ratio σ and prediction entropy (entropy of each pixels in the predictions) in Fig. 12 (right). In the left figure, when the pixel ratio decreases from 95% to 75%, the pixel probability increases rapidly and approximately

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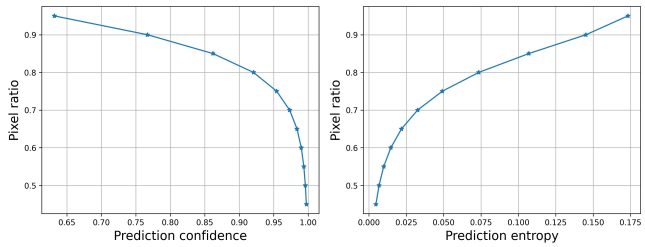

Fig. 12. Relationships between the prediction confidence and pixel ratio (left), and the prediction entropy and pixel ratio (right).

TABLE X

INFLUENCE OF PARAMETER ρ FOR THE SOFTMAX PROBABILITY-BASED SSL METHODS ON THE POTSDAM AND VAIHINGEN DATASETS

Source: Potsdam, Target: Vaihingen								
ρ	40%	50%	60%	70%	75%	80%	90%	100%
mIoU	36.5	37.1	37.4	37.7	**37.9**	37.6	37.2	36.7

TABLE XI

INFLUENCE OF PARAMETER σ FOR THE ESL METHODS ON THE POTSDAM AND VAIHINGEN DATASETS

Source: Potsdam, Target: Vaihingen								
σ	40%	50%	60%	65%	70%	80%	90%	100%
mIoU	35.8	37.5	37.9	**38.1**	38.0	37.7	37.4	36.7

linearly. But, when the pixel ratio decreases from 75% to 40%, the pixel probability increases slowly. Based on this rule, we select the turning point 75% as the parameter ρ to balance the confidence and quantity of the pseudolabels. In the same way, we select the turning point 65% as the parameter σ to balance the entropy and quantity of the pseudolabels.

To validate the aforesaid analysis and further study the influence of the parameters ρ and σ, we train the target classifier utilizing different ρ and σ values via SSL and report the quantitative segmentation results (mIoU) in Tables X and XI. For parameter ρ, we can see that the best performance is achieved when $\rho = 75\%$. Also, the segmentation results are relatively bad when ρ is too high and too low. When ρ is greater than 75%, the incorrect predictions would influence the model in the self-training process. When ρ is less than 75%, the classifier suffers from lack of training samples, which degrades the performance of the segmentation model. For parameter σ, similar results and rules can be found in Table XI, which verifies our analysis.

6) Drawback Discussion: Although our proposed IterDANet achieves the highest m$F1$ score and mIoU values in all experiments, it still has a few short shortcomings and suffers from the rare categories, e.g., car and background. As shown in Table II, when classifying Vaihingen, AdvEnt, BDL, FSDAN, and IntraDANet are better than IterDANet in the class of background. Similarly, when transferring from Vaihingen to Potsdam, AdaptSegNet(m), AdvEnt, BDL, FSDAN, and IntraDANet achieve higher $F1$-score and IoU value in the class of car, and IterDANet performs worse than CycleGAN + SSL, AdaptSegNet, AdvEnt, BDL, FSDAN, and IntraDANet for background category, as illustrated in Table III. For the rare categories, the inter-domain classifier lacks sufficient training samples to extract features. So, the segmentation model usually predicts unsatisfactory maps for the rare classes. After performing the PLGS, the number of training samples of the rare classes is further decreased, which makes the classifier more difficulty to learn the features of the rare classes. On the other hand, iterative training and self training would aggravate the class imbalance problem, which makes the segmentation model perform even worse on the rare classes.

Another drawback is the large workload of model training (e.g., parameter tuning and training time). Due to the two-stage training strategy and plentiful loss functions, our method needs more workload of model training than one-stage methods, which needs to be solved when applying our method in real applications. One possible solution is to use the automatic weighted loss technique, which can learn and automatically adjust the weightings for all the hyperparameter so as to reduce manual labor for the hyperparameter tuning.

V. CONCLUSION

In this article, we propose a three-step iterative IntraDA framework named IterDANet to further reduce the complex intra-domain gap in remote sensing images. First, when conducting the InterDA based on I2I, to alleviate the instability and mode collapse problems of the GAN-based models, we propose a novel GSS to assess all intermediate generators and choose the well-trained generator, which produces high-quality transformed images for the inter-domain segmentation model and improves the performance of the InterDA module. Second, the ER module is designed to generate pseudolabels and rank the target images based on the entropy of predictions; to produce more accurate pseudolabels, a new PLGS is proposed to take advantages of the combination of softmax probability and entropy as the confidence assessment and filter out both low-confident and high-entropy pixels in the predicted maps. Finally, to deal with the complex and diverse distribution shift in the remote sensing images acquired from the same domain, we propose to divide the target domain into multiple subdomains using ER and iteratively align the cleanest subdomain with other noisy subdomains via adversarial learning. The extensive experimental results on the open-source benchmark dataset, which includes aerial images acquired from various cities, validate the superiority of our IterDANet against the existing UDA frameworks, and the performance gap between the UDA framework and supervised learning methods can be further reduced by our IterDANet. Furthermore, extensive ablation studies are conducted to verify the effectiveness of the components (i.e., GSS, PLGS, and iterative IntraDA algorithm), which are proposed to further enhance our IterDANet. In the future work, we will optimize the IntraDA module to make our framework more robust for the class-imbalance problems and further improve the segmentation accuracy of rare categories. Besides, the large workload of model training will be handled to make our methods more practical. Moreover, people will annotate a small number of target images to improve the performance of domain adaptation in practical applications, and our framework will be extended to semisupervised domain adaptation tasks to make our approach play more important role.

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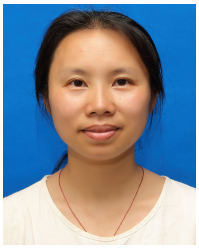

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